

C-DAC Mumbai

Lab Assignment

Question 1: Print Numbers from 1 to N

```
import java.util.Scanner;
```

```
class PrintNumber{  
    public static void main(String args[]){  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter a number : ");  
        int N = sc.nextInt();  
  
        for(int i = 1;i<=N;i++){  
            System.out.print(i + " ");  
        }  
    }  
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac PrintNumber.java  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java PrintNumber  
Enter a number : 10  
1 2 3 4 5 6 7 8 9 10  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java PrintNumber  
Enter a number : 15  
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 2: Print Multiples of 3 between 1 and N

```
import java.util.Scanner;
```

```
class Multipleof3{  
    public static void main(String args[]){  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter a number : ");  
        int N = sc.nextInt();  
  
        for(int i = 1;i<=N;i++){  
            if( i % 3 == 0)  
                System.out.print(i + " ");  
        }  
    }  
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Multipleof3.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Multipleof3
Enter a number : 20
3 6 9 12 15 18
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Multipleof3
Enter a number : 30
3 6 9 12 15 18 21 24 27 30
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>
```

Question 3: Calculate the Factorial of a Number

```
import java.util.Scanner;
```

```
class Factorial{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number : ");
        int N = sc.nextInt();
        int fact = 1;

        for(int i = 1;i<=N;i++){
            fact = fact * i;
        }
        System.out.print("Factorial of "+ N + " is " + fact);
    }
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Factorial.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Factorial
Enter a number : 5
Factorial of 5 is 120
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Factorial
Enter a number : 10
Factorial of 10 is 3628800
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>
```

Question 4: Print Even Numbers from 1 to N

```
import java.util.Scanner;
```

```
class EvenNumbers{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number : ");
        int N = sc.nextInt();

        for(int i = 1;i<=N;i++){
```

```

        if(i % 2 == 0)
            System.out.print(i + " ");
    }
}
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac EvenNumbers.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java EvenNumbers
Enter a number : 10
2 4 6 8 10
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java EvenNumbers
Enter a number : 20
2 4 6 8 10 12 14 16 18 20
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|

```

Question 5: Sum of Odd Numbers between 1 and N

```

import java.util.Scanner;

class SumofOdd{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number : ");
        int N = sc.nextInt();
        int sum = 0;

        for(int i = 1;i<=N;i++){
            if( i % 2 != 0){
                sum = sum + i;
            }
        }
        System.out.print("The sum of odd numbers from 1 to " + N + " is "+ sum);
    }
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac SumofOdd.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java SumofOdd
Enter a number : 10
The sum of odd numbers from 1 to 10 is 25
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java SumofOdd
Enter a number : 20
The sum of odd numbers from 1 to 20 is 100
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|

```

Question 6: Print All Elements of an Array

```

import java.util.Scanner;

```

```

public class ElementsOfArr{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        int[] a = new int[5];

        System.out.print("Enter 5 integers: ");

        for(int i = 0; i < 5; i++){
            a[i] = sc.nextInt();
        }

        for(int i : a){
            System.out.print(i + " ");
        }
    }
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac ElementsOfArr.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java ElementsOfArr
Enter 5 integers: 3 7 12 5 8
3 7 12 5 8
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java ElementsOfArr
Enter 5 integers: 10 15 20 25 30
10 15 20 25 30
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|

```

Question 7: Find the Sum of All Elements in an Array

```
import java.util.Scanner;
```

```

public class SumOfElements {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] a = new int[5];

        System.out.print("Enter 5 integers: ");

        for(int i = 0; i < 5; i++){
            a[i] = sc.nextInt();
        }

        int sum = 0;
        for (int i : a) {
            sum = sum + i;
        }
        System.out.println("The sum of all numbers is : " + sum);
    }
}

```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac SumOfElements.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java SumOfElements
Enter 5 integers: 4 6 8 2 10
The sum of all numbers is : 30

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java SumOfElements
Enter 5 integers: 1 2 3 4 5
The sum of all numbers is : 15

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 8: Print All Names in a String Array

```
import java.util.Scanner;
```

```
public class AllNames {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String[] a = new String[4];

        System.out.print("Enter 4 names: ");
        for(int i = 0; i < 4; i++){
            a[i] = sc.next();
        }

        for(String n : a){
            System.out.println(n + " ");
        }
    }
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac AllNames.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java AllNames
Enter 4 names: Vaishali Avinash Naik Airoli
Vaishali
Avinash
Naik
Airoli

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java AllNames
Enter 4 names: Aditya Manoj Shilbhushan Shweta
Aditya
Manoj
Shilbhushan
Shweta

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 9: Find the Largest Element in an Array

```
import java.util.Scanner;
```

```
import java.util.Arrays;

public class LargestElement {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        int arr[] = new int[5];

        System.out.print("Enter 5 integers: ");

        for(int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }

        Arrays.sort(arr);

        System.out.println("The largest element is: " + arr[4]);
    }
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac LargestElement.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java LargestElement
Enter 5 integers: 12 45 67 23 89
The largest element is: 89

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java LargestElement
Enter 5 integers: 50 20 60 40 1
The largest element is: 60

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 10: Find the Average of Elements in an Array

```
import java.util.Scanner;

public class AvgElement {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        int arr[] = new int[5];

        System.out.print("Enter 5 integers: ");

        for(int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }
    }
}
```

```

int sum = 0;

for(int i = 0; i < 5; i++) {
    sum = sum + arr[i];
}

int avg = sum / 5;

System.out.println("The average of the numbers is: " + avg);
}
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac AvgElement.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java AvgElement
Enter 5 integers: 10 20 30 40 50
The average of the numbers is: 30

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java AvgElement
Enter 5 integers: 1 2 3 4 5
The average of the numbers is: 3

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|

```

Question 11: Count Positive and Negative Numbers in an Array

```

import java.util.Scanner;

public class PosNeg {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        int arr[] = new int[6];

        System.out.print("Enter 6 integers:");

        for(int i = 0; i < 6; i++) {
            arr[i] = sc.nextInt();
        }

        int positive = 0;
        int negative = 0;

        for(int i = 0; i < 6; i++) {
            if(arr[i] > 0) {
                positive++;
            }
            else if(arr[i] < 0) {

```

```

        negative++;
    }
}

System.out.println("Positive numbers: " + positive);
System.out.println("Negative numbers: " + negative);
}
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac PosNeg.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java PosNeg
Enter 6 integers:-5 3 7 -2 0 8
Positive numbers: 3
Negative numbers: 2

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java PosNeg
Enter 6 integers:1 5 2 6 0 0
Positive numbers: 4
Negative numbers: 0

```

Question 12: Sort an Array in Ascending Order

```

import java.util.Scanner;
import java.util.Arrays;

public class SortArray {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        int arr[] = new int[5];

        System.out.print("Enter 5 integers: ");

        for(int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }

        Arrays.sort(arr);

        System.out.print("Sorted array: ");

        for(int i = 0; i < 5; i++) {
            System.out.print(arr[i] + " ");
        }
    }
}

```



```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac SortArray.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java SortArray
Enter 5 integers: 12 45 23 8 90
Sorted array: 8 12 23 45 90
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java SortArray
Enter 5 integers: 56 36 26 86 46
Sorted array: 26 36 46 56 86
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 13: Check if an Array Contains a Specific Element

```
import java.util.Scanner;
```

```
import java.util.Arrays;
```

```
public class SearchArray {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        Integer arr[] = new Integer[5];

        System.out.print("Enter 5 integers : ");

        for(int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.print("Enter the number to search : ");
        int num = sc.nextInt();

        if(Arrays.asList(arr).contains(num)) {
            System.out.println("Found");
        }
        else {
            System.out.println("Not Found");
        }
    }
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac SearchArray.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java SearchArray
Enter 5 integers : 10 20 30 40 50
Enter the number to search : 30
Found

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java SearchArray
Enter 5 integers : 10 20 30 40 50
Enter the number to search : 12
Not Found

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 14: Find the Index of an Element in an Array

```
import java.util.Scanner;
import java.util.Arrays;

public class IndexArray {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        int arr[] = new int[5];

        System.out.print("Enter 5 integers : ");

        for(int i = 0; i < 5; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.print("Enter the number to search : ");
        int num = sc.nextInt();

        int index = Arrays.binarySearch(arr, num);

        if(index >= 0) {
            System.out.println("The number " + num + " is found at index " + index);
        }
        else {
            System.out.println("Not Found");
        }
    }
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac IndexArray.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java IndexArray
Enter 5 integers : 5 10 15 20 15
Enter the number to search : 15
The number 15 is found at index 2

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java IndexArray
Enter 5 integers : 1 2 3 4 5
Enter the number to search : 6
Not Found

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 15: Write a program to print the following pattern

1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2

```
public class Q15 {  
    public static void main(String args[]) {  
  
        for(int i = 1; i <= 5; i++) {  
  
            for(int j = 1; j <= i; j++) {  
                System.out.print(i);  
  
                if(j < i) {  
                    System.out.print("*");  
                }  
            }  
  
            System.out.println();  
        }  
  
        for(int i = 5; i >= 1; i--) {  
  
            for(int j = 1; j <= i; j++) {  
                System.out.print(i);  
  
                if(j < i) {  
                    System.out.print("*");  
                }  
            }  
  
            System.out.println();  
        }  
    }  
}
```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Q15.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Q15
1
2*2
3*3*3
4*4*4*4
5*5*5*5*5
5*5*5*5*5
4*4*4*4
3*3*3
2*2
1
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|

```

Question 16: Write a program to print the following pattern

```

1
1*2
1*2*3
1*2*3*4
1*2*3*4*5

```

```

class Q16{
    public static void main(String[] args) {
        for(int i = 1; i <= 5; i++){
            for(int j = 1; j <= i; j++){
                System.out.print(j);
                if(j < i){
                    System.out.print("*");
                }
            }
            System.out.println();
        }
    }
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Q16.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Q16
1
1*2
1*2*3
1*2*3*4
1*2*3*4*5
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|

```

Question 17: Write a program to print the following pattern

1
1*3
1*3*5
1*3*5*7
1*3*5*7*9

```
public class Q17 {  
    public static void main(String[] args) {  
        for(int i = 1; i <= 5; i++){  
            for(int j = 1; j <= i; j++){  
                System.out.print(2 * j - 1);  
                if(j < i){  
                    System.out.print("*");  
                }  
            }  
            System.out.println();  
        }  
    }  
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Q17.java  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Q17  
1  
1*3  
1*3*5  
1*3*5*7  
1*3*5*7*9  
  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 18: Write a program to print the following pattern

11111
22222
33333
44444
55555

```
public class Q18 {  
    public static void main(String[] args) {  
        for(int i = 1; i <= 5; i++){  
            for(int j = 1; j <= 5; j++){  
                System.out.print(i);  
            }  
            System.out.println();  
        }  
    }  
}
```

```
}  
}  
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Q18.java  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Q18  
11111  
22222  
33333  
44444  
55555  
  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 19: Write a program to print the following pattern

```
1  
22  
333  
4444  
55555
```

```
public class Q19 {  
    public static void main(String args[]) {  
  
        for(int i = 1; i <= 5; i++) {  
  
            for(int j = 1; j <= i; j++) {  
                System.out.print(i);  
            }  
  
            System.out.println();  
        }  
    }  
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Q19.java  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Q19  
1  
22  
333  
4444  
55555  
  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 20: Write a program to print the following pattern

```
1
12
123
1234
12345
```

```
public class Q20 {
    public static void main(String args[]) {

        for(int i = 1; i <= 5; i++) {

            for(int j = 1; j <= i; j++) {
                System.out.print(j);
            }

            System.out.println();
        }
    }
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Q20.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Q20
1
12
123
1234
12345
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>
```

Question 21: Write a program to print the following pattern

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

```
public class Q21 {
    public static void main(String args[]) {

        int num = 1;
```

```

for(int i = 1; i <= 5; i++) {

    for(int j = 1; j <= i; j++) {
        System.out.print(num + " ");
        num++;
    }

    System.out.println();
}
}
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Q21.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Q21
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>

```

Question 22: Write a program to print the following pattern

```

*****
*       *
*       *
*       *
*       *
*       *
*****

```

```

public class Q22 {
    public static void main(String args[]) {

        for(int i = 1; i <= 5; i++) {

            for(int j = 1; j <= 6; j++) {

                if(i == 1 || i == 5 || j == 1 || j == 6) {
                    System.out.print("*");
                }
                else {
                    System.out.print(" ");
                }
            }
        }
    }
}

```



```

        System.out.println();
    }
}
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Q22.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Q22
*****
*      *
*      *
*      *
*****
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|

```

Question 23: Write a program to print the following pattern

```

      *
     ***
    *****
   *********
  ***********
 *****
*****
*****
*****
***
*

```

```

public class Q23 {
    public static void main(String args[]) {

        for(int i = 1; i <= 6; i++) {

            for(int j = 1; j <= 6 - i; j++) {
                System.out.print(" ");
            }

            for(int j = 1; j <= 2 * i - 1; j++) {
                System.out.print("*");
            }

            System.out.println();
        }

        for(int i = 5; i >= 1; i--) {

            for(int j = 1; j <= 6 - i; j++) {

```

```

        System.out.print(" ");
    }

    for(int j = 1; j <= 2 * i - 1; j++) {
        System.out.print("*");
    }

    System.out.println();
}
}
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Q23.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Q23
    *
   ***
  *****
 *****
*****
*****
*****
 *****
  *****
   *****
    ***
     *

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>

```

Question 24: Reverse a String

```

import java.util.Scanner;

public class ReverseString {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String reverse = "";

        for(int i = str.length() - 1; i >= 0; i--) {
            reverse = reverse + str.charAt(i);
        }

        System.out.println("Reversed string: " + reverse);
    }
}

```

```

}

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac ReverseString.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java ReverseString
Enter a string: hello
Reversed string: olleh

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java ReverseString
Enter a string: vaishali
Reversed string: ilahsiav

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|

```

Question 25: Count Vowels in a String

```

import java.util.Scanner;

public class CountVowels {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        int count = 0;

        for(int i = 0; i < str.length(); i++) {
            char ch = str.charAt(i);

            if(ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                count++;
            }
        }

        System.out.println("The number of vowels in '" + str + "' is: " + count);
    }
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac CountVowels.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java CountVowels
Enter a string: programming
The number of vowels in 'programming' is: 3

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java CountVowels
Enter a string: vaishali
The number of vowels in 'vaishali' is: 4

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|

```

Question 26: Check if a String is a Palindrome

```

import java.util.Scanner;

public class Palindrome {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a string: ");
        String str = sc.nextLine();

        String reverse = "";

        for(int i = str.length() - 1; i >= 0; i--) {
            reverse = reverse + str.charAt(i);
        }

        if(str.equals(reverse)) {
            System.out.println("The string '" + str + "' is a palindrome.");
        }
        else {
            System.out.println("The string '" + str + "' is not a palindrome.");
        }
    }
}

```

```

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac Palindrome.java

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Palindrome
Enter a string: madam
The string 'madam' is a palindrome.

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java Palindrome
Enter a string: vaishali
The string 'vaishali' is not a palindrome.

C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|

```

Question 27: String Literal and Object Creation

```

public class StringLiteral {
    public static void main(String args[]) {

        String str1 = "hello";
        String str2 = "hello";

        System.out.println("Both variables point to the same object: " + (str1 == str2));
    }
}

```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac StringLiteral.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java StringLiteral
Both variables point to the same object: true
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 28: String Creation with new Keyword

```
public class StringCreation {
    public static void main(String args[]) {

        String str1 = new String("hello");
        String str2 = new String("hello");

        System.out.println("Using == : " + (str1 == str2));
        System.out.println("Using .equals(): " + str1.equals(str2));
    }
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac StringCreation.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java StringCreation
Using == : false
Using .equals(): true
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 29: String Concatenation and Object Creation

```
public class StringConcatenation {
    public static void main(String args[]) {

        String str1 = "hello";
        String str2 = "world";
        String str3 = str1 + str2;

        System.out.println("Is str3 pointing to the same object as str1? " + (str3 == str1));
    }
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac StringConcatenation.java
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java StringConcatenation
Is str3 pointing to the same object as str1? false
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 30: String Pool with intern() Method

```
public class StringPool {  
    public static void main(String args[]) {  
  
        String str1 = new String("hello");  
        String str2 = str1.intern();  
        String str3 = "hello";  
  
        System.out.println("Is str2 and str3 pointing to the same object? " + (str2 == str3));  
    }  
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac StringPool.java  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java StringPool  
Is str2 and str3 pointing to the same object? true  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```

Question 31: Multiple String Literals with Same Content

```
public class StringLiterals {  
    public static void main(String args[]) {  
  
        String str1 = "java";  
        String str2 = "java";  
        String str3 = "java";  
  
        System.out.println("All strings point to the same object: " + (str1 == str2 && str2 == str3));  
    }  
}
```

```
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>javac StringLiterals.java  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>java StringLiterals  
All strings point to the same object: true  
C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 4\Assignment>|
```